Using Calculators to Solve Inequalities

Exam code: 0607

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Using Calculators to Solve Inequalities

How do I solve inequalities using my graphic display calculator?

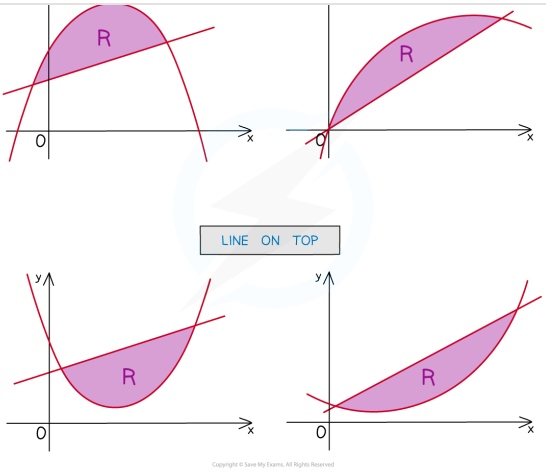
- You can use a **graphic display calculator** to **solve inequalities** of the form
  - $f_1(x) > f_2(x)$ or $f_1(x) < f_2(x)$
  - $f_1(x) \geq f_2(x)$ or $f_1(x) \leq f_2(x)$
- First, turn the inequality into an **equation** and **solve** $f_1(x) = f_2(x)$
  - Read the revision note on Using Calculators to Solve Equations to see how
  - The **solutions** to the equation $f_1(x) = f_2(x)$ are the ***x*-coordinates** of the **points of intersection** of the two graphs $y = f_1(x)$ and $y = f_2(x)$
- Then look at the regions (areas) enclosed between the two graphs and decide which curve is the **top curve** and which is the **bottom curve**
  - e.g. The **solution** to $f_1(x) > f_2(x)$ are the **ranges of *x*-values** for which $y = f_1(x)$ is the top curve and $y = f_2(x)$ is the bottom curve
    - The top curve is **greater than** the bottom curve
  - Examples of different regions, R, are shown below

Help me understand

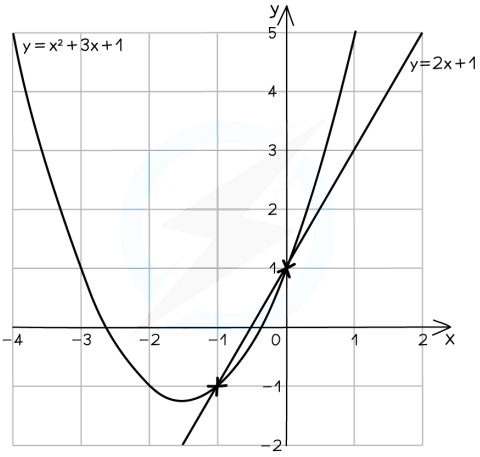
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



- For example, to solve $x^2 + 3x + 1 > 2x + 1$ using a graphical method
  - First solve the "=" equation to get $x = -1$ and $x = 0$
  - Then plot on your calculator and look for any regions where the quadratic is above (**greater than**) the line (quadratic on top, line on bottom)
  - This happens to the left of $x = -1$ and to the right of $x = 0$
  - The solution is $x < -1$ or $x > 0$



Examiner Tips and Tricks

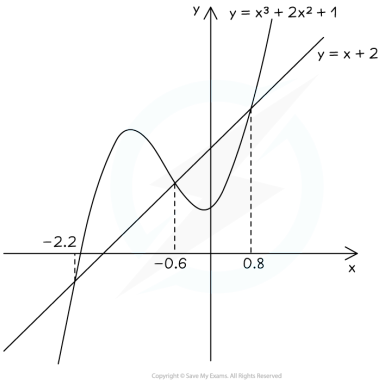
Remember to give your solutions as **ranges** of *x* values, using strict inequalities if the question uses strict inequalities!

Worked Example

Start a new graph on your calculator and enter $f_1(x) = x^3 + 2x^2 + 1$ to generate the graph
Add on the second graph of $f_2(x) = x + 2$

Select the option to "analyze" the graph (this may be labelled as G-Solv)
Choose the option "intersections" to find the coordinates of the 3 points of intersection

The solutions to the equation are the x-coordinates of the points of intersection



The solutions to the equation $x^3 + 2x^3 + 1 = x + 2$ are
$x = -2.2, x = -0.6$ or $x = 0.8$ to 1 d.p.

The inequality in the question is "cubic graph $\leq$ straight line graph"
Look at regions where the cubic is the bottom curve and the line is above

left of $x = -2.2$ and between $x = -0.6$ and $x = 0.8$

The solutions are the ranges of x-values over which these regions lie
Write your final answer using non-strict inequalities (to match the question)

$x \leq -2.2$ or $-0.6 \leq x \leq 0.8$ to 1 d.p.

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